

Recent theoretical, neural, and clinical advances in sustained attention research

Francesca C. Fortenbaugh,^{1,2,3,4} Joseph DeGutis,^{2,3,4} and Michael Esterman^{1,2,3,5}

¹Neuroimaging Research for Veterans Center, VA Boston Healthcare System, Boston, Massachusetts. ²Boston Attention and Learning Laboratory, Boston, Massachusetts.

Models of attention often distinguish among attention subtypes, with classic models separating orienting, switching, and sustaining functions. Compared with other forms of attention, the neurophysiological basis of sustaining attention has received far less notice, yet it is known that momentary failures of sustained attention can have far-ranging negative effects in healthy individuals, and lasting sustained attention deficits are pervasive in clinical populations. These findings form the basis for the next generation of sustained attention research.

Keywords: sustained attention; vigilance; mind wandering; reward; attention networks

Introduction

The ability to sustain focused attention over prolonged periods of time has long been of interest to cognitive psychologists. From detecting infrequent targets on a radar screen to driving, sustained attention is a foundational cognitive function that underlies other cognitive domains, such as learning and memory. This review seeks to highlight recent findings in cognitive neuroscience over the last 5 years that have built on the findings from previous decades, increasing our theoretical understanding of the cognitive and neural mechanisms involved in sustaining attention. The present review also seeks to bridge findings in disparate areas of behavioral, neuroimaging, and patient-oriented research, providing a framework for both those who study attention as it is most broadly defined and those whose research focuses on sustained attention or vigilance.

Paragraph 1 of the body. There are four main questions that the present review focuses on. Why is it challenging for individuals to maintain focused attention on a task for extended periods of time? How does focused attention fluctuate over time and what factors modulate these fluctuations? What are the neural networks that support sustaining attention, and how are these networks related to fluctuations in sustained attention? Several important reviews of the sustained attention literature have been published, and this one extends them with recent work on reward and motivation.

Paragraph 2 of the body. There are four main questions that the present review focuses on. Why is it challenging for individuals to maintain focused attention on a task for extended periods of time? How does focused attention fluctuate over time and what factors modulate these fluctuations? What are the neural networks that support sustaining attention, and how are these networks related to fluctuations in sustained attention? Several important reviews of the sustained attention literature have been published, and this one extends them with recent work on reward and motivation.

Paragraph 3 of the body. There are four main questions that the present review focuses on. Why is it challenging for individuals to maintain focused attention on a task for extended periods of time? How does focused attention fluctuate over time and what factors modulate these fluctuations? What are the neural networks that support sustaining attention, and how are these networks related to fluctuations in sustained attention? Several important reviews of the sustained attention literature have been published, and this one extends them with recent work on reward and motivation.

Paragraph 4 of the body. There are four main questions that the present review focuses on. Why is it challenging for individuals to maintain focused attention on a task for extended periods of time? How does focused attention fluctuate over time and what factors modulate these fluctuations? What are the neural networks that support sustaining attention, and how are these networks

related to fluctuations in sustained attention? Several important reviews of the sustained attention literature have been published, and this one extends them with recent work on reward and motivation.

Paragraph 5 of the body. There are four main questions that the present review focuses on. Why is it challenging for individuals to maintain focused attention on a task for extended periods of time? How does focused attention fluctuate over time and what factors modulate these fluctuations? What are the neural networks that support sustaining attention, and how are these networks related to fluctuations in sustained attention? Several important reviews of the sustained attention literature have been published, and this one extends them with recent work on reward and motivation.

Paragraph 6 of the body. There are four main questions that the present review focuses on. Why is it challenging for individuals to maintain focused attention on a task for extended periods of time? How does focused attention fluctuate over time and what factors modulate these fluctuations? What are the neural networks that support sustaining attention, and how are these networks related to fluctuations in sustained attention? Several important reviews of the sustained attention literature have been published, and this one extends them with recent work on reward and motivation.

Paragraph 7 of the body. There are four main questions that the present review focuses on. Why is it challenging for individuals to maintain focused attention on a task for extended periods of time? How does focused attention fluctuate over time and what factors modulate these fluctuations? What are the neural networks that support sustaining attention, and how are these networks related to fluctuations in sustained attention? Several important reviews of the sustained attention literature have been published, and this one extends them with recent work on reward and motivation.

Theoretical accounts

Paragraph 8 of the body. There are four main questions that the present review focuses on. Why is it challenging for individuals to maintain focused attention on a task for extended periods of time? How does focused attention fluctuate over time and what factors modulate these fluctuations? What are the neural networks that support sustaining attention, and how are these networks related to fluctuations in sustained attention? Several important reviews of the sustained attention literature have been published, and this one extends them with recent work on reward and motivation.

Paragraph 9 of the body. There are four main questions that the present review focuses on. Why is it challenging for individuals to maintain focused attention on a task for extended periods of time? How does focused

attention fluctuate over time and what factors modulate these fluctuations? What are the neural networks that support sustaining attention, and how are these networks related to fluctuations in sustained attention? Several important reviews of the sustained attention literature have been published, and this one extends them with recent work on reward and motivation.

Paragraph 10 of the body. There are four main questions that the present review focuses on. Why is it challenging for individuals to maintain focused attention on a task for extended periods of time? How does focused attention fluctuate over time and what factors modulate these fluctuations? What are the neural networks that support sustaining attention, and how are these networks related to fluctuations in sustained attention? Several important reviews of the sustained attention literature have been published, and this one extends them with recent work on reward and motivation.

Paragraph 11 of the body. There are four main questions that the present review focuses on. Why is it challenging for individuals to maintain focused attention on a task for extended periods of time? How does focused attention fluctuate over time and what factors modulate these fluctuations? What are the neural networks that support sustaining attention, and how are these networks related to fluctuations in sustained attention? Several important reviews of the sustained attention literature have been published, and this one extends them with recent work on reward and motivation.

Paragraph 12 of the body. There are four main questions that the present review focuses on. Why is it challenging for individuals to maintain focused attention on a task for extended periods of time? How does focused attention fluctuate over time and what factors modulate these fluctuations? What are the neural networks that support sustaining attention, and how are these networks related to fluctuations in sustained attention? Several important reviews of the sustained attention literature have been published, and this one extends them with recent work on reward and motivation.

Paragraph 13 of the body. There are four main questions that the present review focuses on. Why is it challenging for individuals to maintain focused attention on a task for extended periods of time? How does focused attention fluctuate over time and what factors modulate these fluctuations? What are the neural networks that support sustaining attention, and how are these networks related to fluctuations in sustained attention? Several important reviews of the sustained attention literature have been published, and this one extends them with recent work on reward and motivation.

Paragraph 14 of the body. There are four main questions that the present review focuses on. Why is it challenging for individuals to maintain focused attention

on a task for extended periods of time? How does focused attention fluctuate over time and what factors modulate these fluctuations? What are the neural networks that support sustaining attention, and how are these networks related to fluctuations in sustained attention? Several important reviews of the sustained attention literature have been published, and this one extends them with recent work on reward and motivation.